

Fiat 124 Spider 2017+ Apple CarPlay / Android Auto Upgrade. Part 1 – Firmware Update



This is part one of a guide covering the steps required to install the 'official' Mazda upgrade to the CMU (what we used to call 'the stereo') to add Apple CarPlay and Android Auto functionality in your 2017+ FIAT or Abarth 124 Spider. ***This part of the guide covers updating the firmware in the CMU and must be carried out prior to steps described in part two.*** The version 70 firmware will run on your existing hardware quite happily so you can, for example, carry out the firmware update one weekend and complete the hardware install the following weekend.

A couple of caveats you should be aware of if you are contemplating this upgrade:

1. This update is only available if you have the 7" display. The 3" displays in some Classics are not compatible.
2. FCA are being their usual useless selves and have not issued a FIAT / Abarth specific firmware version for this upgrade. This means we need to use a Mazda firmware for now. This has three main consequences:
 - a. The startup and splash-screens and various other parts of the interface will reference Mazda rather than FIAT / Abarth (note the 'Mazda' icon in the screenshot above for example). Thankfully, this can mostly be corrected via Ameridan's fantastic tweaks – we'll cover this later.

- b. After the firmware update, the Bluetooth ID for the car changes to “Mazda”. Again, this can be fixed, but for now, just be aware you will need to re-pair your phone with the car following the update.
 - c. The in-built navigation will stop working. The system does a lookup of your vehicle’s VIN as part of its security checks. Mazda firmware expects to see a Mazda VIN, FIAT firmware, a FIAT VIN. Because our cars present a FIAT VIN to Mazda firmware after the upgrade, the in-built NAV no longer works. As with the other issues, there is a workaround for this that we’ll cover, but you’ll also have access to Apple and Google maps, Waze etc after the upgrade so you may not care.
- 3. If you are taking your car in for service, do not let the dealership attempt to update your CMU firmware - nothing good will come of it. There is a FIAT specific firmware version that is newer than the one the 2017 and early 2018 cars came with so it is possible they will try – however that will break your CarPlay / Android Auto compatibility.

WHAT YOU WILL NEED:

To update the firmware, you will need:

- V70.00.100a Mazda firmware.
- A blank FAT32 formatted USB stick – 8 to 16GB is recommended and please use a quality name-brand stick rather than the cheapest piece-of-you-know-what you can find. If the stick fails on you during the firmware flash, you could end up with a non-functional CMU.
- Ameridan's ID7 Tweak Package
- The Mazda to Fiat V70 Universal Tweak Package

The download locations for the firmware files and each of two tweak packages will be covered later.

USEFUL LINKS AND ACKNOWLEDGEMENTS

The information in this how-to has been pulled from a number of sources and I hereby nod my hat to:

Ameridan and his fantastic blog and work on the All-In-Tweaks at <https://21stcenturyfiat124spider.wordpress.com> This is well worth checking out even if you never have any intention of doing the CarPlay / Android Auto update. Ameridan is also responsible for many of the tweaks and ideas in the Universal tweak package.

ASH8 for his how-to thread at <https://mazda3revolution.com/forums/2014-2018-mazda-3-skyactiv-audio-electronics/230249-genuine-carplay-android-auto-all-here-now.html>

I was inspired by rsmagee and TheSalamander – Fiat 124 CarPlay pioneers – in this thread: <https://www.124spider.org/forum/26-electronics-audio-lighting/30666-carplay-up-running.html>

The original Mazda All-In-One tweaks are at <https://mazdatweaks.com/>

ATTENTION – IT IS ESSENTIAL THAT THE FIRMWARE UPGRADE DESCRIBED HERE IS CARRIED OUT BEFORE THE INSTALLATION OF ANY OF THE NEW HARDWARE DESCRIBED IN PART 2.

In order for Apple CarPlay and Android Auto to be available, we have to upgrade our CMUs to run the Mazda version 70 firmware (firmware is simply the software that controls the unit). Our Fiats and Abarths shipped with a Fiat / Abarth branded derivative of either version 56 (2017 and early 2018 cars) or version 59 (later 2018 and 2109 cars).

The version 56 firmware allowed tweaking of certain features via software developed by a group of enthusiasts - including things like enabling the touchscreen while the car is moving, changing background images and swapping FIAT logos for the Abarth Scorpion. Mazda, and presumably FIAT's, lawyers got nervous about this as they considered some of tweaks to be bypassing safety features (obviously none of them had ever tried to use the command wheel to enter an address). As a result, starting with version 59 of the firmware Mazda blocked the ability to apply tweaks. FIAT started shipping 124's with their derivative of version 59 firmware mid-way through the 2018 model run.

Not to be beaten, the very clever developers of the tweak software developed a patch (called ID7) that allows the tweaks to continue to be installed on version 59 and later firmware, ***but only if the patch is installed BEFORE upgrading to version 59 firmware or later.*** As we'll be upgrading to version 70 firmware, we need to install the patch before upgrading to retain the ability to apply tweaks. It is also this patch that will allow us to restore the built-in navigation function following the upgrade.

As my car is a 2017 and shipped with the version 56 firmware, I was able to install the patch using the method described below. If you have a 2018 or 2019 with the version 59 firmware already installed (or your dealer updated you to it during a service), there is an alternative method of installing the patch described at <https://mazdatweaks.com/serial/> It's a more involved, but currently it's your only option. I haven't tried this process and the instructions below assume you are starting from version 56 firmware. You can check your firmware version on the CMU by going to:

HOME -> SETTINGS -> SYSTEM -> ABOUT -> VERSION INFORMATION



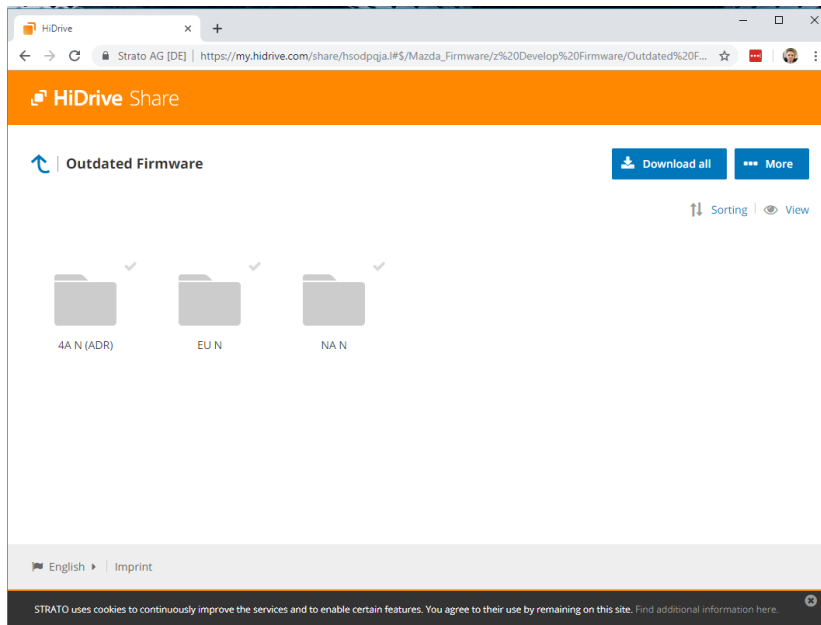


STEP 1 – DOWNLOAD REQUIRED FILES

We'll start by downloading the required firmware files and two tweak packages.

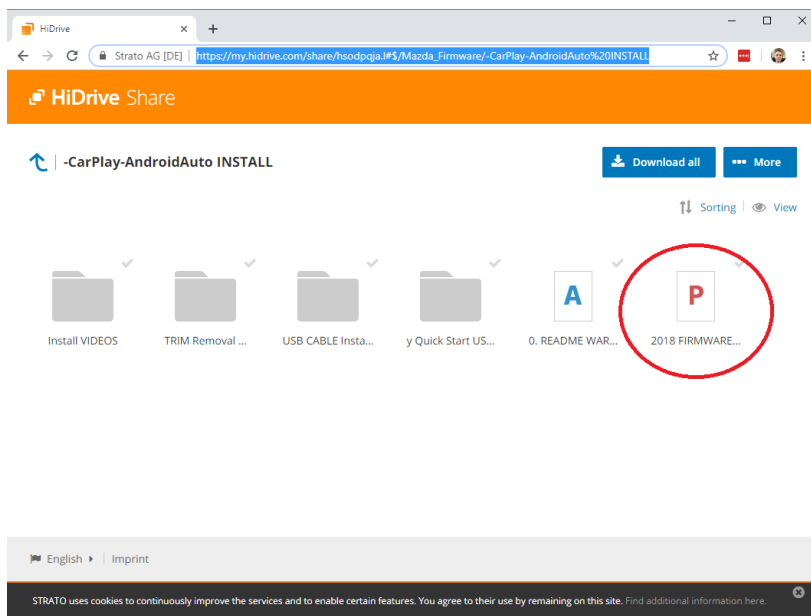
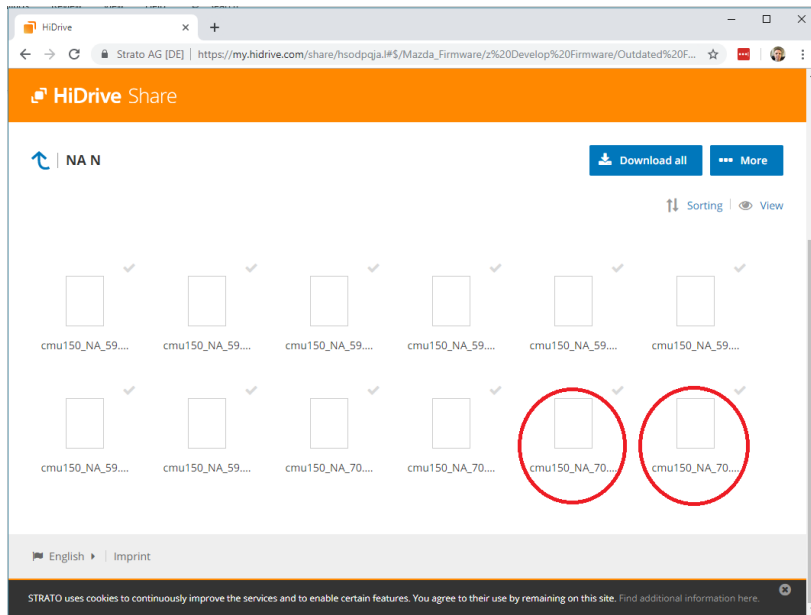
1. Go to

[https://my.hidrive.com/share/hsodpqja.l#\\$/Mazda_Firmware/z%20Develop%20Firmware/Outdated%20Firmware](https://my.hidrive.com/share/hsodpqja.l#$/Mazda_Firmware/z%20Develop%20Firmware/Outdated%20Firmware) to get the required firmware files. Note the necessary files are in the “Outdated Firmware” folder. As of May 2019, there is a newer Mazda firmware version than the one we will use. **The newer firmware *SHOULD NOT* be used.** Pay attention to the file names in the following steps to ensure you have the right versions.



2. Go to the folder relevant to your region of the world; NA N for North America, EU N for Europe, 4A N for Oceania. The following steps assume you've gone to the NA N folder – just swap in the corresponding files for other regions as appropriate.
3. Download the two version 70 files and the Mazda instructions PDF to your computer:
 - a. cmu150_NA_70.00.100A_failsafe.up
 - b. cmu150_NA_70.00.100A_reinstall.up
4. Download the Mazda instructions PDF (2018 FIRMWARE CMU UPDATE PROCEDURE-WORLDWIDE.pdf from [https://my.hidrive.com/share/hsodpqja.l#\\$/Mazda_Firmware/-CarPlay-AndroidAuto%20INSTALL](https://my.hidrive.com/share/hsodpqja.l#$/Mazda_Firmware/-CarPlay-AndroidAuto%20INSTALL)

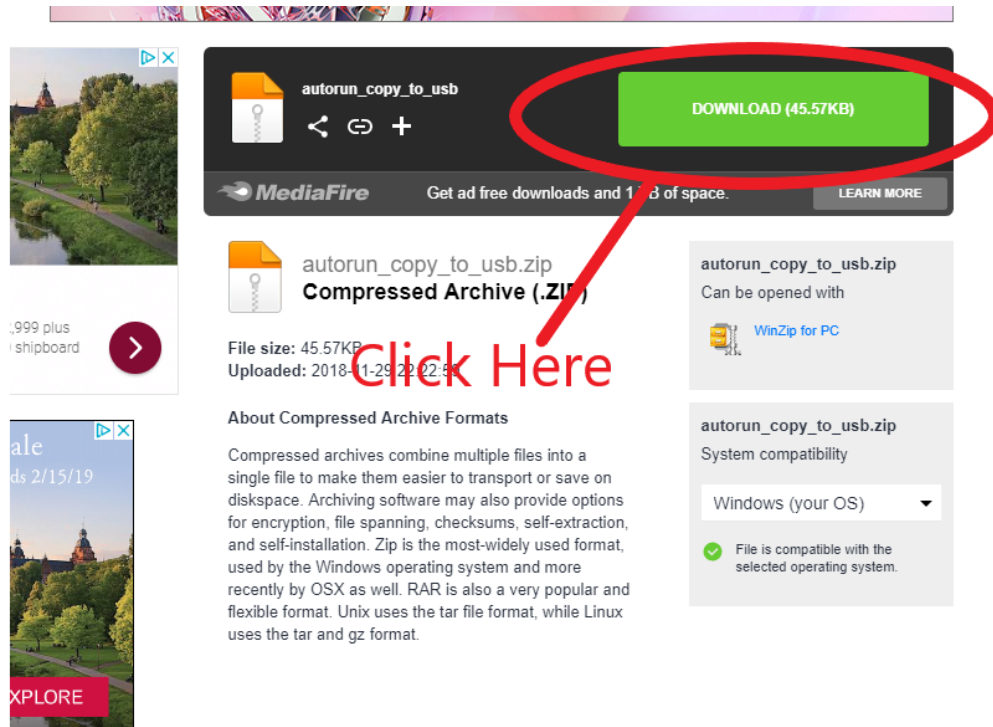
Download the files one at a time rather than use the 'download all files' options as the folder method compresses everything into a zip file which can corrupt the update ('up') files.



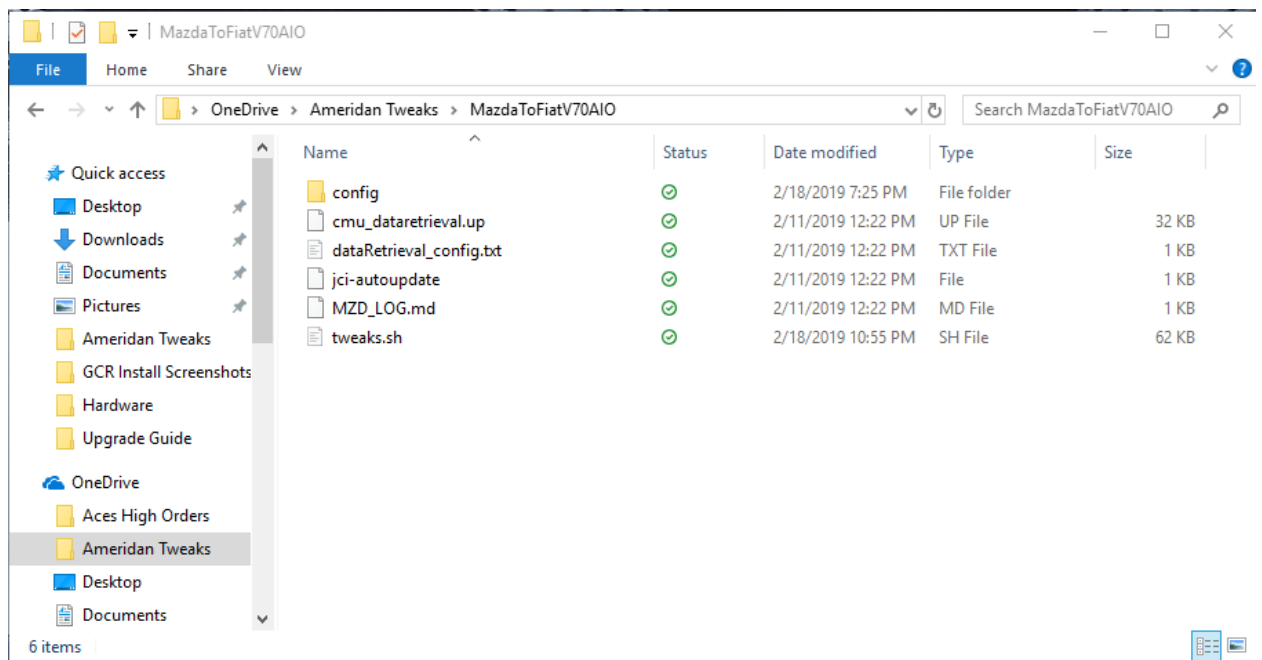
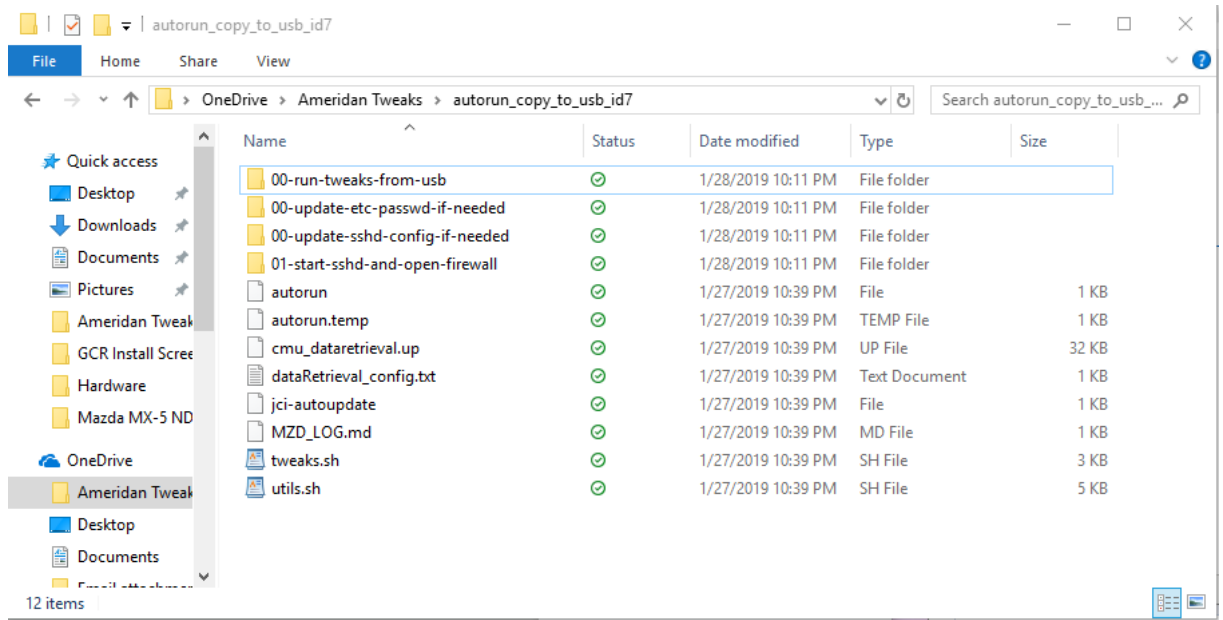
If you are OK with Mazda branded splash screens and backgrounds and have no desire to install any of the tweaks described on Ameridan's blog or restoring the built in Navigation, go straight to the Mazda version 70 firmware installation (step 5) Otherwise, continue with all steps below.

5. Download Ameridan's ID7 Tweak Package, and the Mazda to Fiat V70 Universal Tweak package from the following URLs respectively:
- a. http://www.mediafire.com/file/0r6pzhongok9h0u/autorun_copy_to_usb.zip
 - b. <http://www.mediafire.com/file/zi0cp690a7x5rq1/MazdaToFiatV70AIO.zip/file>

At each URL, click the large 'download button':

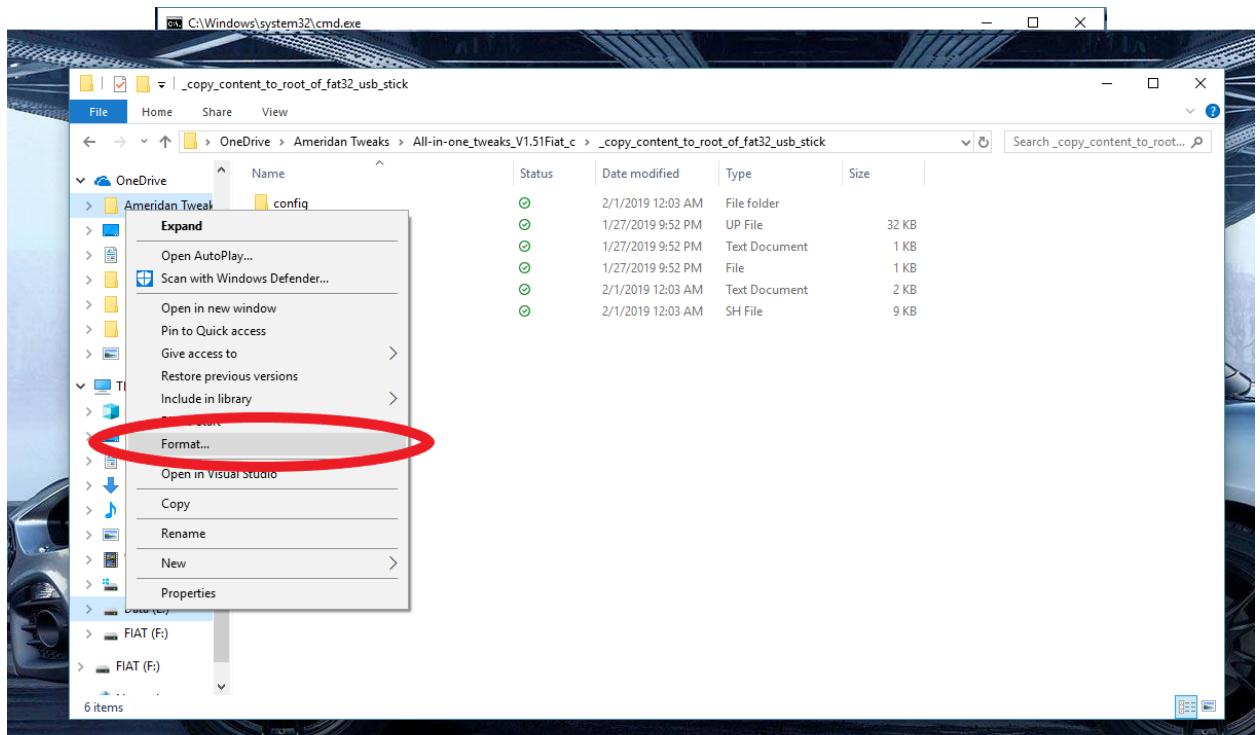


6. Unzip the contents of the zip files to folders on your computer. I put them in folders called Ameridan Tweaks/autorun_copy_to_usb_id7, and Ameridan Tweaks/MazdaToFiatV70AIO respectively:

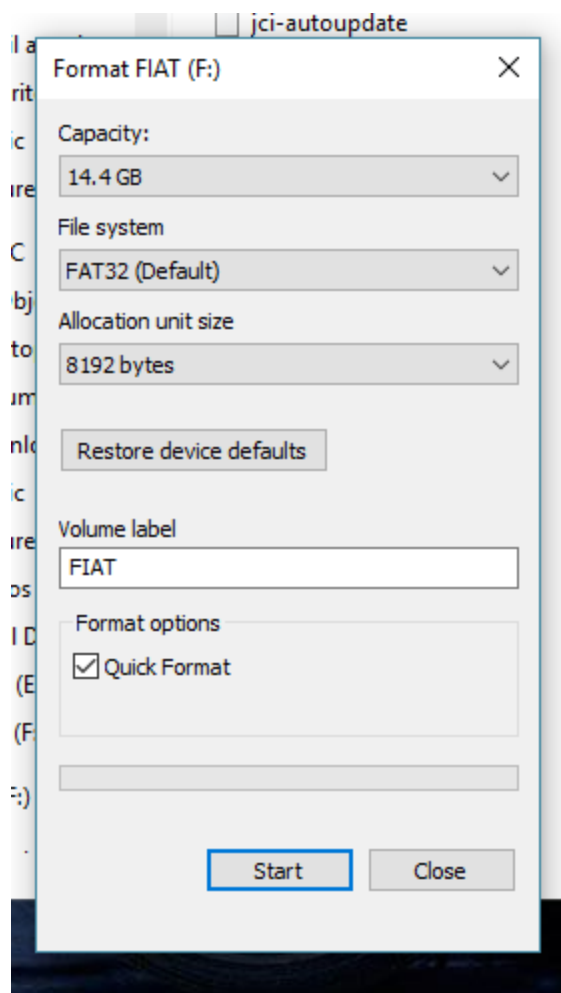


STEP 2 – FORMAT YOUR USB DRIVE

1. Plug your USB stick in to your computer and back up any files on it you want to keep for later – we are about to wipe it so you will lose any existing files you don't back up.
2. Format your USB stick as FAT32 – to do this, right click on the drive in Windows Explorer and select "Format":



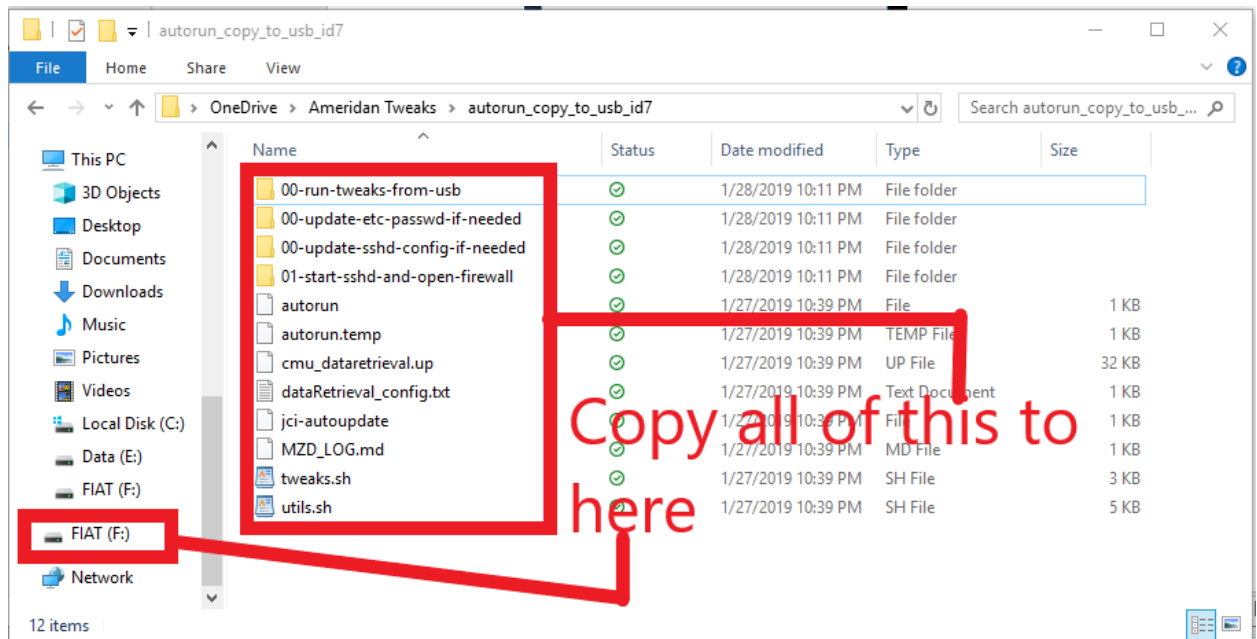
3. In the resulting screen, make sure the file system is set to FAT32 and enter a volume label (this is the label used to describe the drive in File Explorer). I set my volume label to "FIAT". Click OK to begin the format and click OK on the warnings about losing any existing data on the drive.



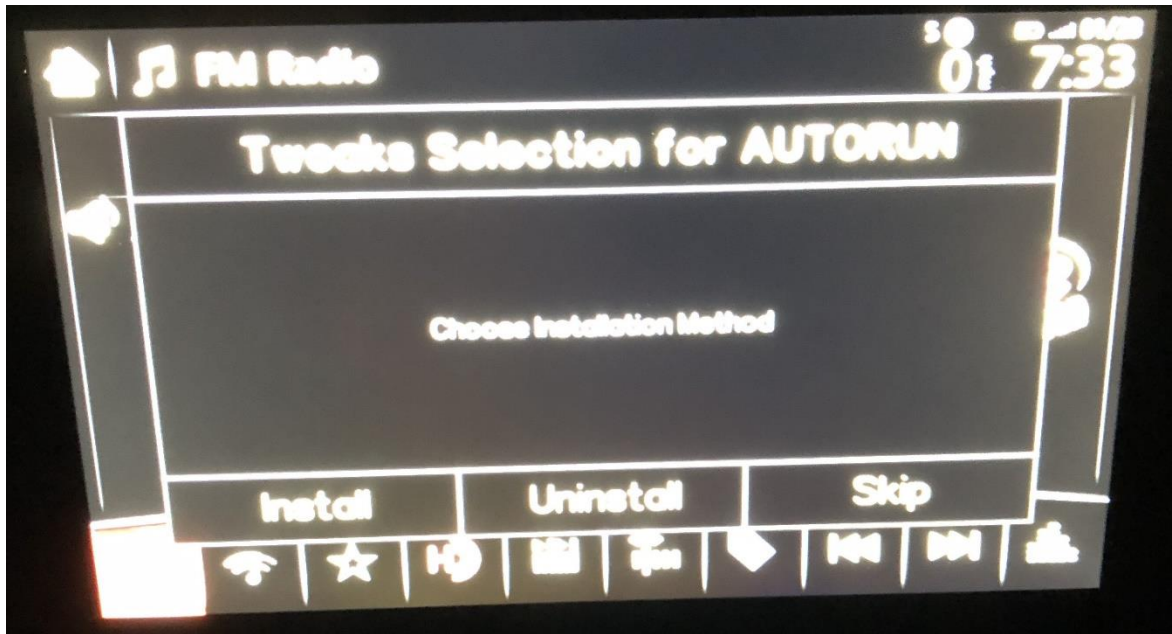
STEP 3 – INSTALL ID7 PATCH

We'll next install the ID7 patch on your CMU. This patch will allow us to replace much of the Mazda branding in the version 70 firmware and restore the built-in Nav function after we complete the firmware upgrade. If your car is currently on a version 59 firmware, you will need to follow the process described at <https://mazdatweaks.com/serial/>. The steps below assume you are on a version 56 firmware.

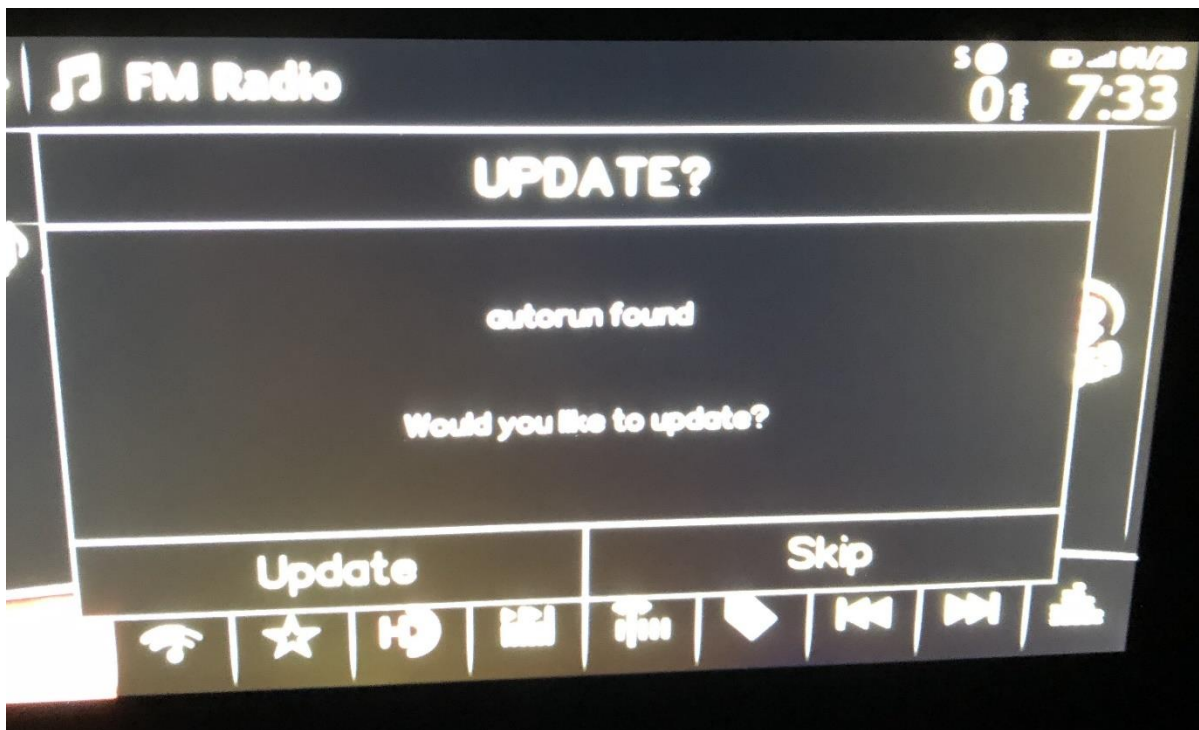
1. Copy the contents of the unzipped ID7 package (in folder "autorun_copy_to_usb_id7" on my computer) to your USB drive.



2. Safely eject your USB stick. We're now ready to go to your car.
3. Turn off Bluetooth on any phones you have paired with your car.
4. Go to your car and unplug any connected USB cables and remove your navigation SD card.
5. Plug in your USB stick.
6. Without touching any of the pedals, hit the start button once to put the car in to accessory mode and turn the CMU on.
7. Switch audio source to FM
8. After a few minutes a screen will pop up asking if you want to select an installation method. Use the touchscreen to click on Install



9. You'll then see a screen asking you to confirm you want to run the update. Click Update.



10. After a minute or two, you'll get a message saying the update is complete and asking if you want to reboot. Click reboot. When the screen goes blank, remove your USB stick.



The ID7 patch is now installed. Go back to your computer.

STEP 4 – INSTALLING THE MAZDA VERSION 70 FIRMWARE.

For the install of the version 70 firmware files, we'll be following the official Mazda instructions. The process is very similar to the ID7 patch install we just completed. They are very clear with good screenshots so I'm not going to try and recreate them.

1. Plug your USB stick in to your computer and delete all existing files from it.
2. Copy the two update files to your USB drive. The filenames below assume you are in North America – use the files appropriate to your region:
 - a. cmu150_NA_70.00.100A_failsafe.up
 - b. cmu150_NA_70.00.100A_reinstall.up
3. Print off the Mazda instructions (2018 FIRMWARE CMU UPDATE PROCEDURE-WORLDWIDE.pdf). Read and familiarize yourself with them. There are thirty steps.
4. Carry out the updates according to the Mazda instructions.

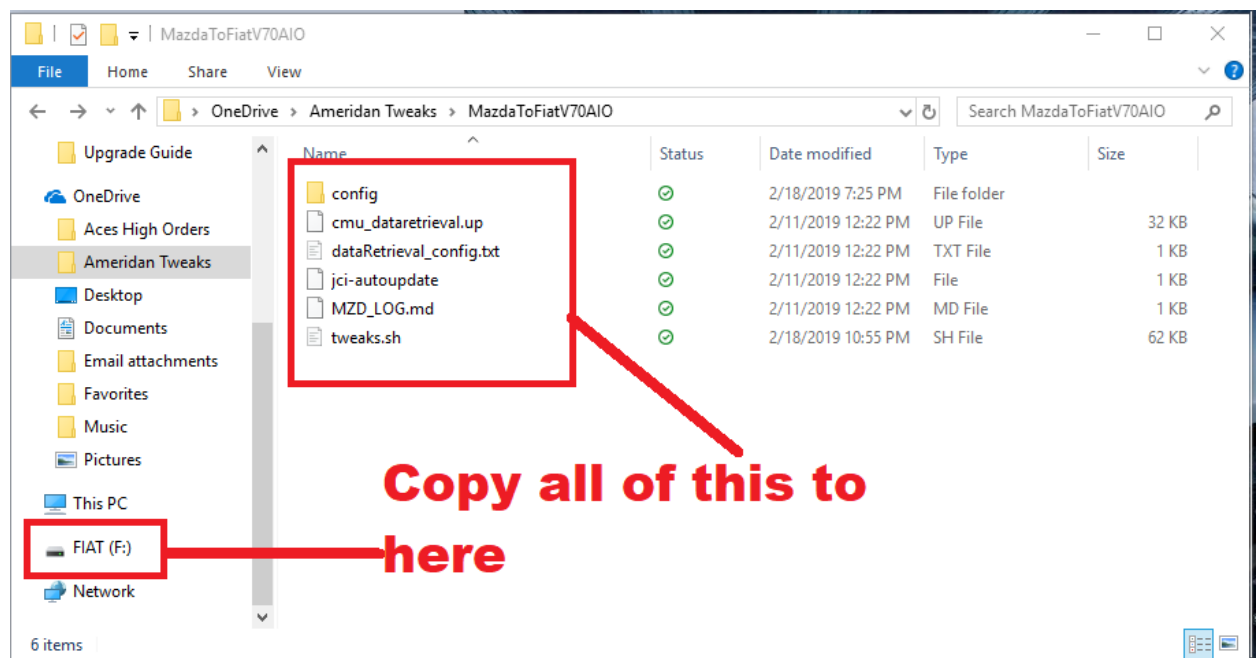
You'll be installing the two files one after the other – the failsafe file followed by the main update file. The overall process will take about an hour – make sure you have enough time to complete it. Pay attention to the note about depressing the brake or clutch every 20 minutes during the install to prevent the CMU shutting down in the middle of the update.

Once the update is complete, we've done everything needed to support CarPlay and Android Auto. The next two steps are optional and only needed if you want to replace the Mazda branding with Fiat or Abarth equivalents or if you want to restore the built in Navigation functionality.

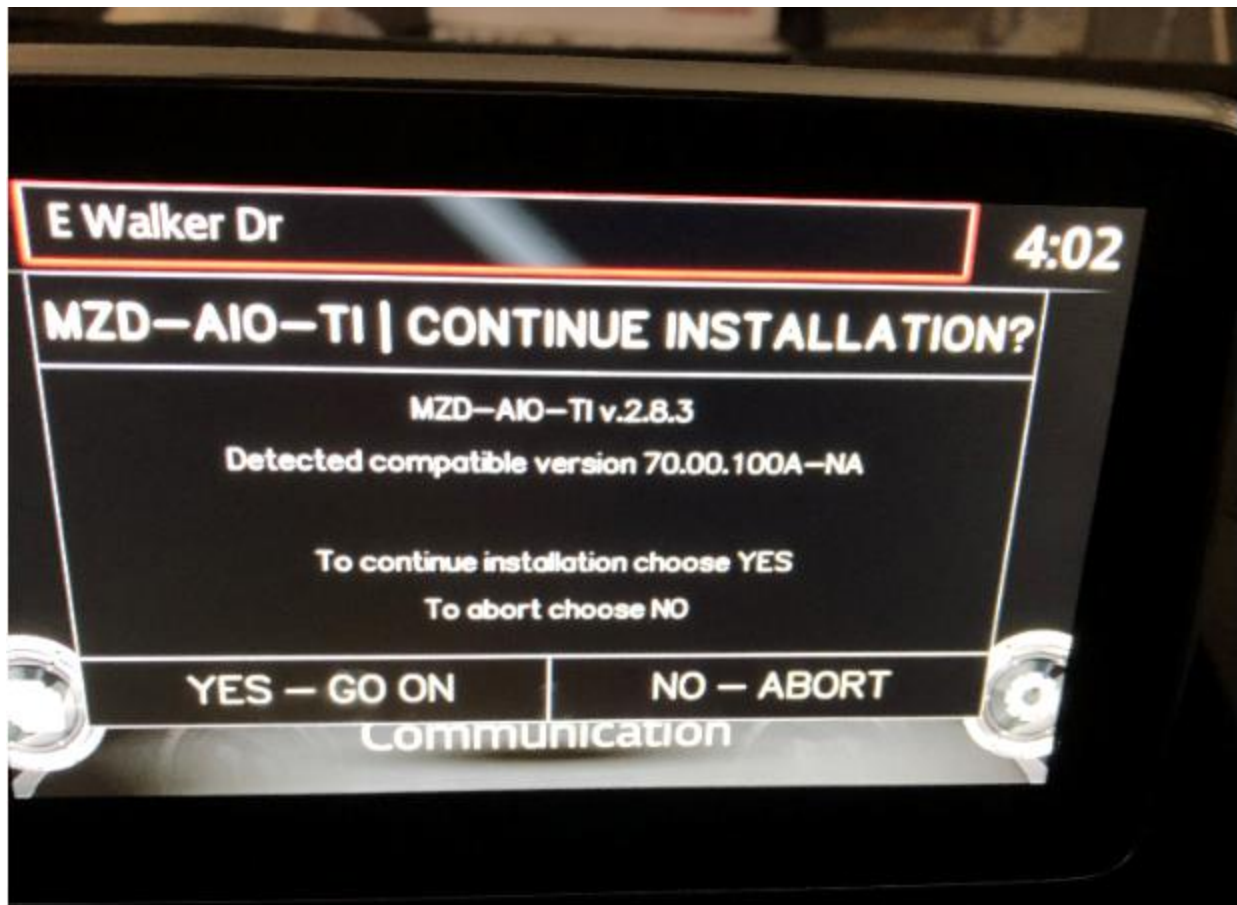
STEP 5 – REPLACE MAZDA BRANDING WITH FIAT / ABARTH EQUIVALENTS AND RESTORING BUILT IN NAVIGATION FUNCTIONALITY

To replace the Mazda startup / shutdown animations that ship with version 70 firmware with Fiat or Abarth equivalents, and to restore the OEM Navigation function, carry out the steps below. If you have previously installed any tweaks using Ameridan's All-in-one installer, in particular the Speedometer application, it is recommended that you uninstall those tweaks prior to running these steps. They can be re-installed once this process is complete:

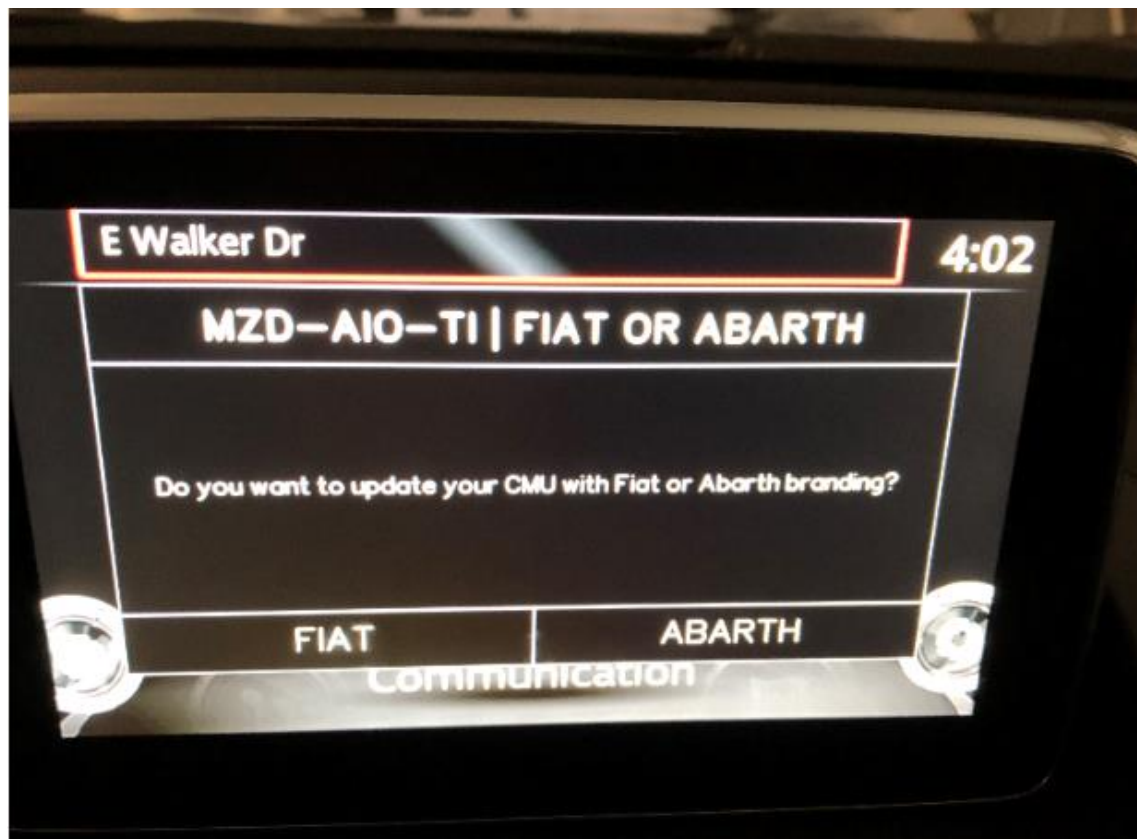
1. Plug your USB stick in to your computer and delete all the files on it.
2. Copy the contents of the unzipped MazdaToFiatV70AIO.zip file (in folder ("Ameridan Tweaks/ MazdaToFiatV70AIO" on my computer) to your USB Drive.



3. Safely eject your USB stick. We're now ready to go back to your car.
4. Unpair your phone from the car in the phone's Bluetooth settings and switch off Bluetooth on your phone.
5. Unplug any USB cables connected to your car and remove your navigation SD card.
6. Without touching any of the pedals, hit the start button once to put the car in to accessory mode and turn the CMU on.
7. Switch audio source to FM
8. In the car's Bluetooth settings screen, unpair all previously paired phones. **IT IS IMPORTANT YOU DO THIS PRIOR TO RUNNING THE TWEAKS**
9. Plug your USB stick in to the car.
10. After a few minutes a screen will pop up asking you to confirm the installation. Click 'YES'.



11. You will now be prompted with a series of questions asking what changes you wish to make. The first will ask if you want to use Fiat or Abarth logos and branding:



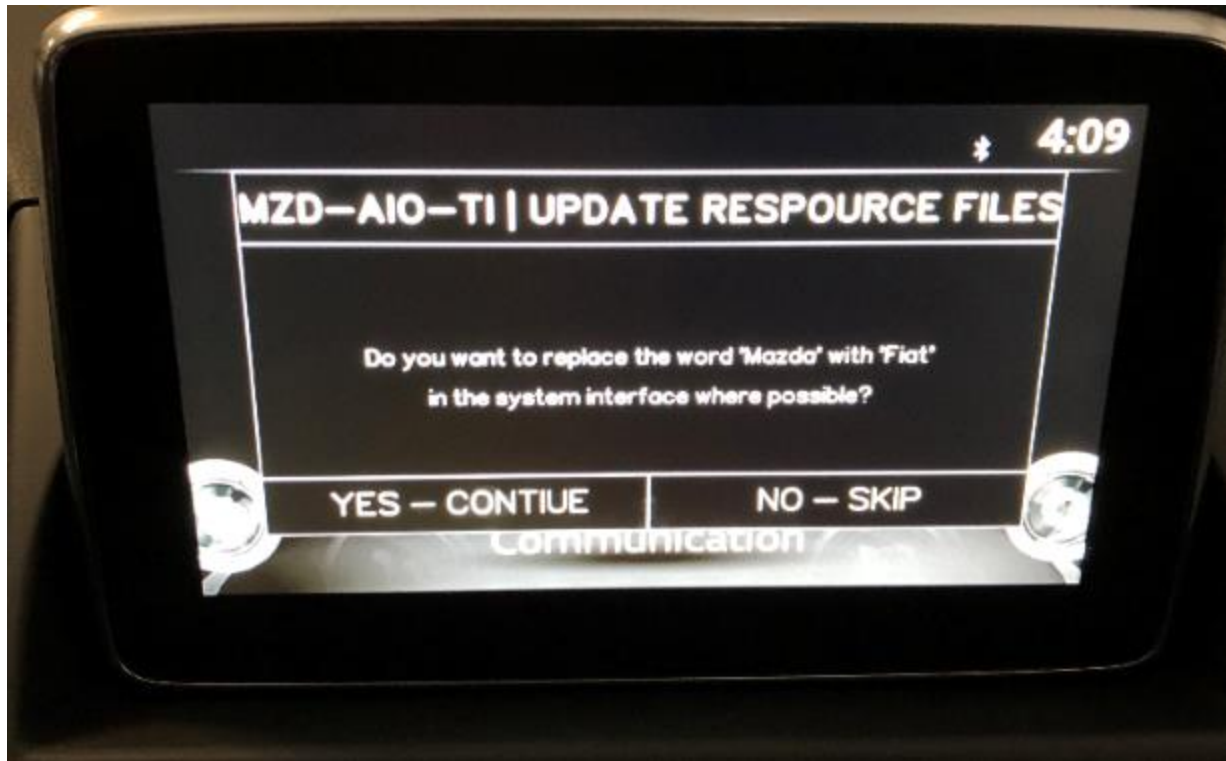
Click on FIAT or ABARTH according to your choice – this will determine whether the later steps install Fiat or Abarth startup and shutdown animations and whether the Mazda CarPlay icon gets replaced with a Fiat or Abarth icon.

12. Next, you will be asked if you want to replace the Mazda startup and shutdown animations:



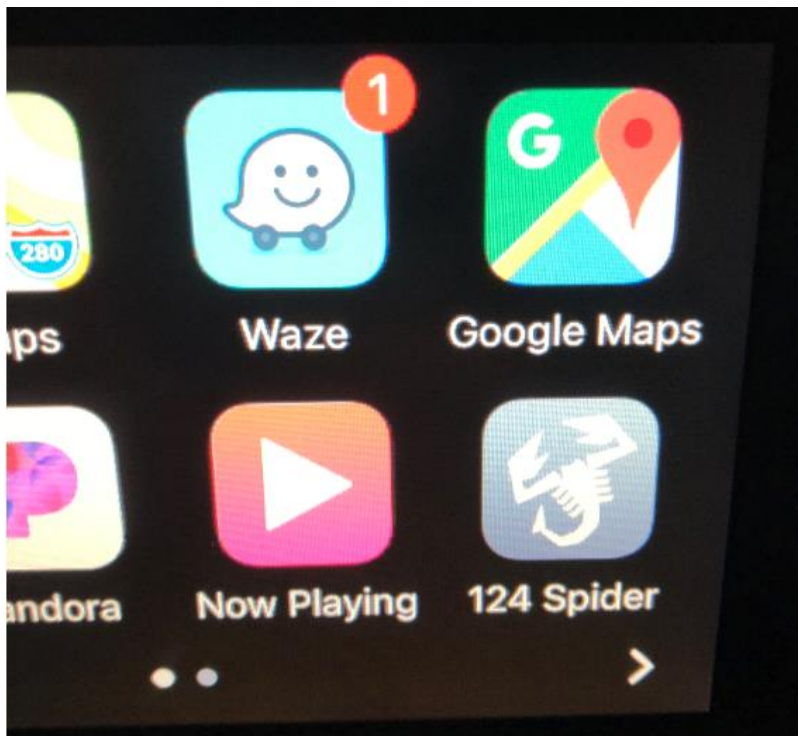
If you answer 'YES', the animations will be updated – this will take a moment to complete. If you answer 'NO', the system will jump to the next tweak. Note that even after installing this change, you may still see a Mazda screen briefly from time to time – particularly if you do a hard reboot of the system.

13. Next you will be asked if you wish to replace the word 'Mazda' with '124 Spider', 'Abarth', or 'Fiat' as appropriate throughout the system interface:

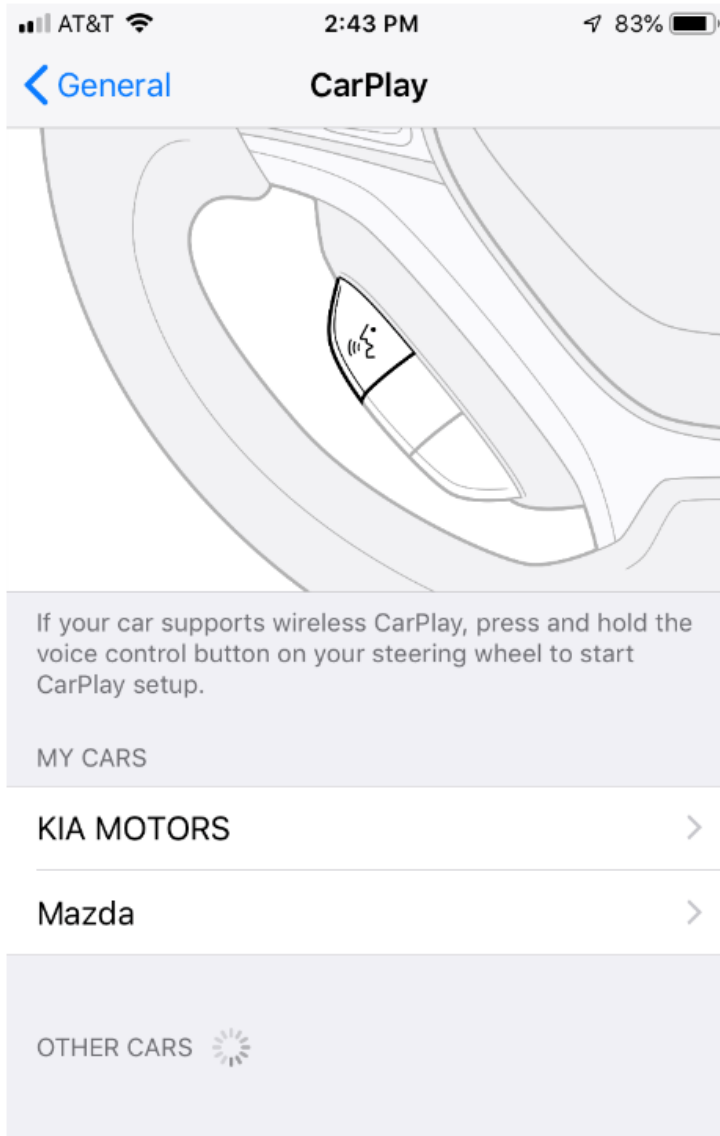


If you answer 'YES', the system interface will be updated – this will take a moment to complete. Note this should work for systems using UK, Australian and North American English, Italian, German, French (including Canadian French), Spanish and Dutch. If you answer 'NO', the system will jump to the next tweak.

14. Next you will be asked if you want to replace the Mazda Icon in Apple CarPlay. This should result in an Abarth Scorpion or Fiat logo icon labelled '124 Spider' being used instead of the Mazda icon in the CarPlay homescreen:



If you answer 'YES', the icon will be updated – this will take a moment to complete. If you answer 'NO', the system will jump to the next tweak. Note that even after applying this change, the car will still be listed as 'Mazda' in your iPhone's CarPlay settings:

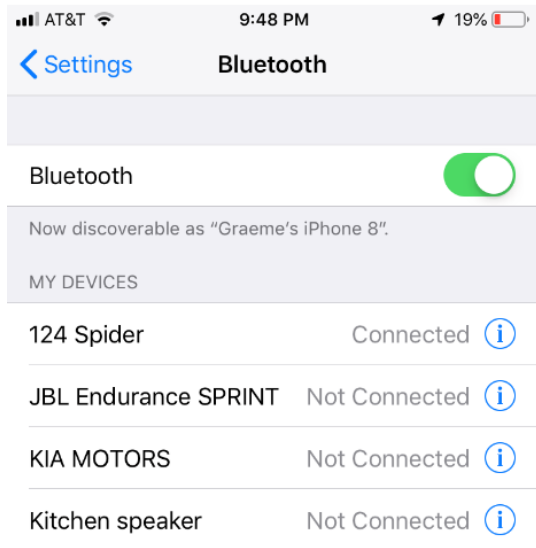


15. Next you will be asked if you wish to restore the OEM navigation functionality:



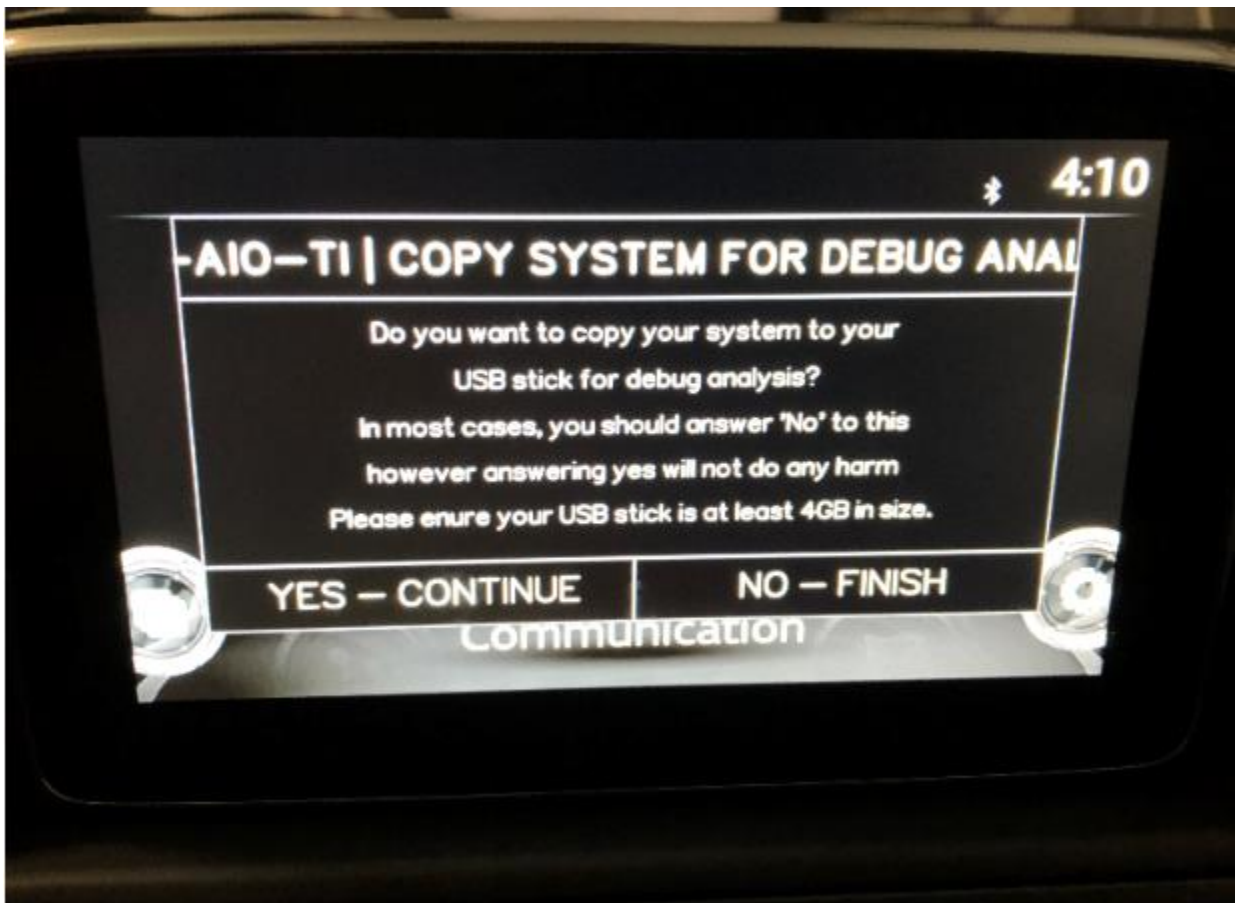
If you answer 'YES', the navigation function will be restored – this will take several minutes to complete. **During this process, depress and release the clutch or brake pedal to prevent the CMU being put in to sleep mode.** If you answer 'NO', the system will jump to the next tweak.

16. Next you will be asked if you want to change the vehicle's Bluetooth ID from 'Mazda' to '124 Spider'. This is the name you will see when pairing your phone with the car:



If you answer 'YES', the Bluetooth ID will be updated – this will take a moment to complete. If you answer 'NO', the system will jump to the next tweak.

17. The final tweak you will be asked about makes a copy of your entire CMU system on your USB stick. In most cases, this tweak is unnecessary and should be skipped, but if you are trying to debug an issue or are simply curious, click yes.

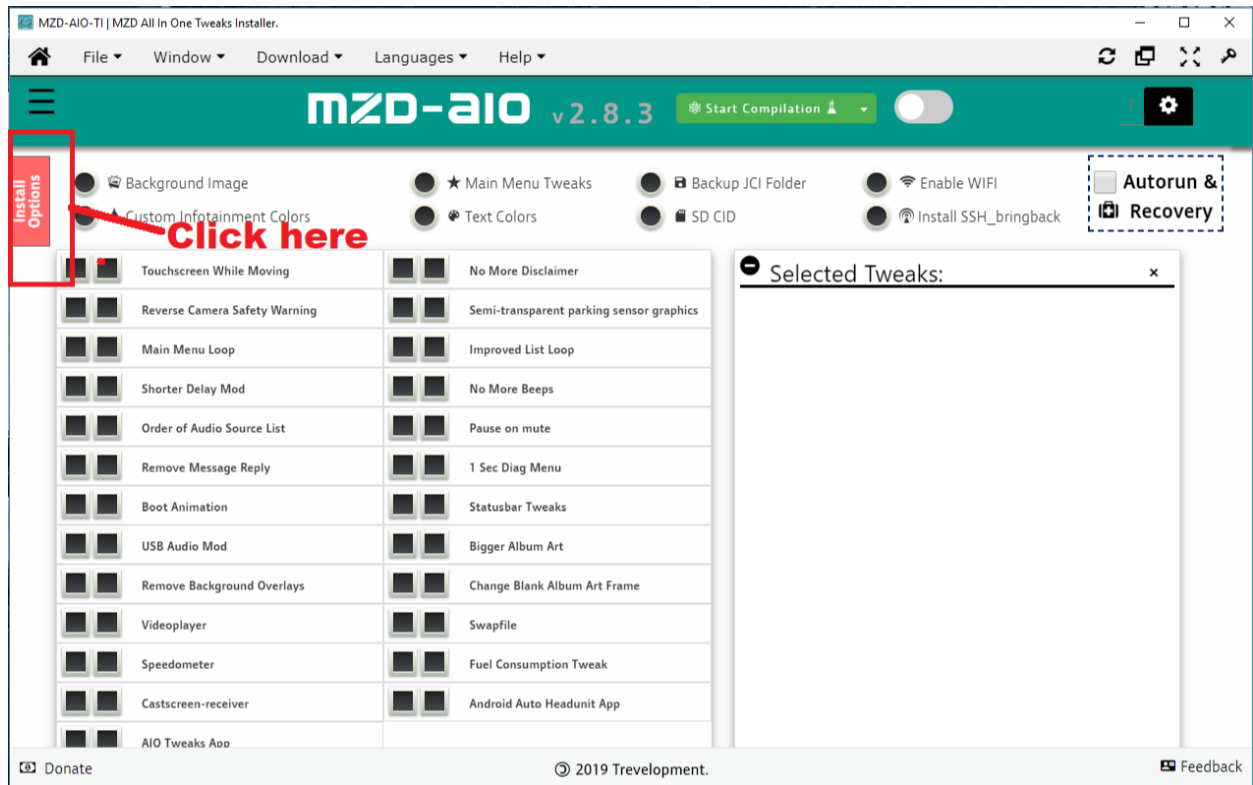


If you answer 'YES', the backup will be created – this will take up to 30 minutes to complete. During this process, depress and release the clutch or brake pedal every 10 minutes to prevent the CMU being put in to sleep mode. The backup will be written to config/full-backup on your USB drive. If you answer 'NO', the tweak installations will be complete.

18. After the last tweak is completed, the CMU will reboot and you will be prompted when it is safe to remove your USB stick. After the reboot, all your selected changes should be available. Remember to re-pair your phone with the car.

The tweaks installer is safe to run as often as you like, and you can pick and choose which tweaks to install on each run.

Following these steps, if you intend to install any further tweaks as described on Ameridan's blog or at www.mazdatweaks.com, it is recommended that you use the All-In-Installer from www.mazdatweaks.com and check the 'Install to Resources' option in the Install Options tab:



The firmware update process is now complete, and you can continue to the hardware installation process described in part 2.

